# Structural Boundaries and Performance Limitations of STL Decomposition Preprocessing in Multi-Step Meteorological Time Series Forecasting

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***Abstract*—This study investigates the structural boundary conditions and empirical efficacy of Seasonal-Trend decomposition using Loess (STL) when deployed as an explicit preprocessing step for multi-step time series forecasting. Utilizing high-frequency daily mean temperature data from Hanoi, Vietnam (2015–2025, $n = 3,653$), we conduct a rigorous symmetrical evaluation between three end-to-end models (SARIMA, XGBoost, LSTM) and three decompose-then-forecast hybrid pipelines (STL+SARIMA, STL+XGBoost, STL+LSTM) across horizons $t+1$, $t+3$, and $t+7$. Counter to conventional theoretical assumptions, empirical results demonstrate severe performance degradation when utilizing STL decomposition. At a short-term horizon ($t+1$), the STL+LSTM pipeline yields a Root Mean Squared Error (RMSE) of $2.8482°$C, representing an 84.35% increase in error compared to the end-to-end LSTM model (RMSE = $1.5450°$C). Symmetrical degradation is verified across all machine learning and deep learning configurations via the Diebold-Mariano test ($p < 0.01$). Diagnostic post-mortem analysis reveals that the core breakdown is driven by the structural properties of the series: an exceptionally low strength of trend ($F_T = 0.0140$) coupled with high seasonal dominance ($F_S = 0.8016$). Under these conditions, the Loess-extracted trend component injects additive noise rather than predictive signal, while restricting non-linear models to model non-stationary residuals induces critical overfitting. We propose a conditional framework establishing quantitative boundaries for decomposition-based time series pipelines.**

***Index Terms*—Time series forecasting, STL decomposition, Long Short-Term Memory, XGBoost, Strength of Trend, Multi-horizon forecasting, Diebold-Mariano test.**

## I. Introduction

Multi-step daily temperature forecasting is essential for grid load balancing, agricultural scheduling, and metropolitan climate resilience planning. Due to the complex, non-linear dynamics governing atmospheric interactions, standard time series methodologies frequently struggle with long-term dependencies and high-frequency variations. A widely adopted paradigm in contemporary forecasting literature is the *decompose-then-forecast* approach, where raw data is systematically separated into isolated structural components prior to optimization.

Seasonal-Trend decomposition using Loess (STL) represents the state-of-the-art in non-parametric decomposition, prized for its robustness against extreme outliers and capability to handle highly non-linear seasonal variations. The fundamental assumption underlying hybrid frameworks is that filtering out deterministic components reduces the mathematical complexity of the data stream, enabling downstream statistical or machine learning architectures to optimize solely on simplified stationary sub-signals. Large-scale forecasting benchmarks, such as the M4 Competition [10], have continuously underscored the potential of hybrid systems, yet a formal characterization of the structural conditions under which decomposition impairs forecasting remains an open research gap.

This paper addresses this gap by utilizing a decade-long daily meteorological dataset from Hanoi, Vietnam (2015–2025). We implement a strict, symmetrical experimental design comparing end-to-end architectures against their STL-preprocessed equivalents.

Our core contributions are threefold: 1) providing clear empirical proof that forcing STL decomposition on an asset with an exceptionally weak long-term trend systematically degrades multi-step accuracy; 2) validating the statistical significance of this degradation across statistical, machine learning, and deep learning paradigms using the Diebold-Mariano framework; and 3) defining quantitative structural boundaries based on component variance metrics to serve as an architectural guardrail for time series engineering.

## II. Mathematical Framework and Methodology

### A. STL Decomposition and Feature Quantification

The additive STL framework isolates the observed time-series target $Y_t$ into trend ($T_t$), seasonal ($S_t$), and remainder ($R_t$) components:

$$Y_t = T_t + S_t + R_t \tag{1}$$

The decomposition is executed via a system of two nested loops. The inner loop updates the seasonal and trend components through alternating steps of subseries smoothing and low-pass filtering utilizing localized polynomial regression (Loess) [3]. The outer loop computes robustness weights $\rho_t$

based on a bisquare function of the remainder $R_t$ to reduce the influence of anomaly spikes:

$$\rho_t = B\left(\frac{|R_t|}{6 \cdot \text{median}(|R_t|)}\right) \tag{2}$$

To mathematically define the information profile of the target series, we adopt the metric proposed by Wang et al. [11] to quantify the strength of trend ($F_T$) and strength of seasonality ($F_S$):

$$F_T = \max\left(0, 1 - \frac{\text{Var}(R_t)}{\text{Var}(T_t + R_t)}\right) \tag{3}$$

$$F_S = \max\left(0, 1 - \frac{\text{Var}(R_t)}{\text{Var}(S_t + R_t)}\right) \tag{4}$$

### B. Predictive Architectures

The experimental matrix comprises six distinct model setups divided into two structural paradigms:

*1) Group A (End-to-End Models):* A1 (SARIMA): Seasonal Autoregressive Integrated Moving Average based on the classical mathematical foundation of Box and Jenkins [1], operating directly on the raw, un-decomposed target.

A2 (XGBoost): A Gradient Boosted Decision Tree architecture formulated by Chen and Guestrin [2], transforming a dense temporal feature matrix directly to $Y_{t+h}$.

A3 (LSTM): A recurrent deep learning network composed of gated memory blocks designed by Hochreiter and Schmidhuber [6] to internally capture sequential context from $Y_t$.

*2) Group B (Decompose-then-Forecast Models):* The Group B pipelines implement a decoupled strategy:

$$\hat{Y}_{t+h} = \hat{T}_{t+h} + S_{\text{pattern}}[\text{doy}(t+h)] + \hat{R}_{t+h} \tag{5}$$

where $S_{\text{pattern}}$ is an extracted deterministic daily seasonal pattern indexed by day-of-year (doy $\in [1, 366]$).

B1 (STL+SARIMA): The combined component ($T_t + R_t$) is modeled via an automated SARIMA structure.

B2 (STL+XGBoost): The high-frequency remainder $R_t$ is forecasted by XGBoost [2], while the trend is modeled via a local linear estimator.

B3 (STL+LSTM): The network [6] is trained exclusively on the bounded sequence of the remainder $R_t$.

## III. Experimental Configuration

### A. Data Characterization and Data Splitting

The empirical data consists of daily historical records from the Hanoi weather station officially curated by the Visual Crossing Corporation database [12], spanning Jan 1, 2015 to Dec 31, 2024 ($n = 3,653$). The primary target is daily mean temperature (temp, °C).

To ensure realistic evaluation free from temporal data leakage, a strict chronological data split is applied. The Training Set spans from 2015-01-01 to 2022-12-31 ($n_{\text{train}} = 2,922$). The Validation Set covers 2023-01-01 to 2023-12-31 ($n_{\text{val}} = 365$). The Test Set encompasses 2024-01-01 to 2024-12-31 ($n_{\text{test}} = 366$).

### B. Hyperparameter Optimization and Setup

The engineering configuration mirrors the exact execution architecture defined in the centralized pipeline.

For the STL Specs, the seasonal period is set to $M = 365$ [3]. Robust Loess optimization is active with linear degrees enforced for both trend and seasonal tracking (trend_deg $= 1$, seasonal_deg $= 1$).

For the SARIMA Optimization, automated order selection utilizes heuristics implemented within the standard algorithmic framework [7], constrained to $\max(p, q) = 3$ and seasonal $\max(P, Q) = 2$. To maintain scalability over multi-step updates, full structural refitting is restricted to a 30-day sliding batch window, with intermittent updates handled recursively.

For the XGBoost Architecture, the model is configured with $n_{\text{estimators}} = 500$, max_depth $= 6$, and learning_rate $= 0.05$ [2]. The input feature space includes temporal lag days $L \in \{1, 2, 3, 7, 14, 30, 365\}$ and rolling statistics calculated across window lengths of 7 and 30 days. This feature engineering pipeline expands the 15 raw meteorological variables into a total input matrix 42 features per time step, providing both short-term lag context and multi-scale rolling statistics to the XGBoost and LSTM architectures.

For the LSTM Specifications, the network is configured as a stacked recurrent architecture [6] with dense layers containing $[64, 32]$ hidden units, standard dropout of $0.2$, and a lookback window fixed at 30 days. Optimization is driven by Adam with early stopping triggered after 10 epochs of validation loss stagnation.

## IV. Empirical Results and Visualization

### A. Structural Diagnostic Findings

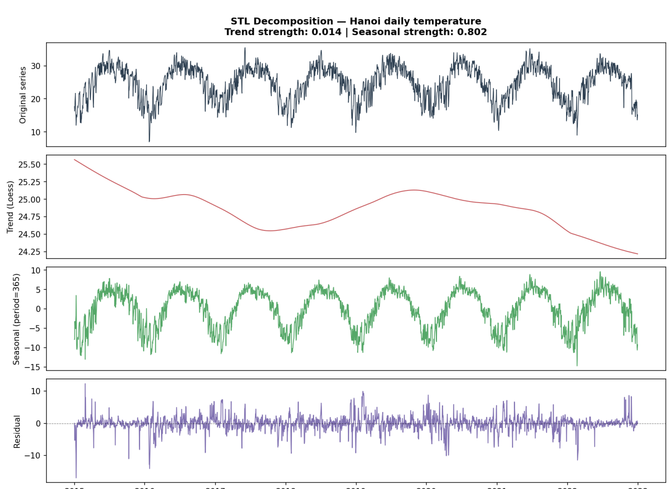


Fig. 1: Visual diagnostics of STL decomposition. The red line (Trend) demonstrates a nearly flat profile ($F_T = 0.014$), confirming that explicit trend modeling injects additive noise into the forecasting pipeline rather than capturing a meaningful signal.

Applying the mathematical variance criteria proposed by Wang et al. [11] to the training partitions yields highly skewed results regarding the data's composition. The Strength of Trend ($F_T$) is computed as $0.0140$, while the Strength of Seasonality ($F_S$) is evaluated at $0.8016$. The baseline data possesses virtually zero linear trend information ($F_T \approx 1.4\%$), whereas the annual seasonal envelope controls over $80.16\%$ of total sample variance.

To evaluate the mathematical properties of the target series prior to model execution, standard non-stationarity diagnostics are implemented on the raw temperature sequence. The Augmented Dickey-Fuller (ADF) test [4], applied with regression specification 'ct' (constant and linear trend) and 19 lags selected by AIC, yields a test statistic of $-3.2142$ and a p-value of $0.0816$, failing to reject the unit root null hypothesis at the conventional 5% significance level. The KPSS test [8] produces a statistic of $0.0406$ with $p > 0.10$, also failing to reject its null hypothesis of trend-stationarity. The conflicting signals from the two complementary tests – ADF failing to confirm non-stationarity while KPSS fails to confirm stationarity – indicate that the series occupies the boundary between these regimes, a characteristic signature of a seasonally integrated process. This structural property provides formal motivation for the seasonal differencing operator ($D = 1$, $s = 365$) employed in the SARIMA specification, which effectively resolves the near-unit-root behavior by differencing across annual cycles. Furthermore, a Ljung-Box test [9] on the raw temperature series rejects the white noise null hypothesis across all evaluated lags $p < 0.001$ in all cases), confirming pervasive autocorrelation consistent with the strong seasonal structure quantified by $FS = 0.8016$.

### B. Symmetrical Model Evaluation and Degradation Plots

Table I summarizes the cross-model error metrics evaluated systematically across horizons $t + 1$, $t + 3$, and $t + 7$.

TABLE I: Comprehensive Forecasting Performance Matrix

| Horizon | Model | RMSE (°C) | MAE (°C) | MAPE (%) | Skill |
|---|---|---|---|---|---|
| **t+1** | SARIMA | 1.7199 | 1.3251 | 5.8025 | -0.0721 |
| | XGBoost | 1.6005 | 1.2081 | 5.2338 | 0.0023 |
| | **LSTM (Best)** | **1.5450** | **1.2032** | **5.3205** | **0.0369** |
| | STL+SARIMA | 2.0538 | 1.5524 | 7.0039 | -0.2802 |
| | STL+XGBoost | 2.8000 | 2.1269 | 9.6795 | -0.7453 |
| | STL+LSTM | 2.8482 | 2.1055 | 9.7284 | -0.7754 |
| **t+3** | SARIMA | 2.3989 | 1.8131 | 8.2410 | 0.1899 |
| | XGBoost | 2.9678 | 2.2254 | 10.1365 | -0.0022 |
| | **LSTM (Best)** | **2.1419** | **1.6464** | **7.5381** | **0.2767** |
| | STL+SARIMA | 2.9569 | 2.1397 | 9.9966 | 0.0015 |
| | STL+XGBoost | 2.8004 | 2.1278 | 9.6846 | 0.0543 |
| | STL+LSTM | 2.9811 | 2.1970 | 10.1634 | -0.0067 |
| **t+7** | SARIMA | 2.7136 | 2.0824 | 9.5897 | 0.2651 |
| | XGBoost | 3.7266 | 2.8079 | 13.1039 | -0.0093 |
| | **LSTM (Best)** | **2.5316** | **1.9289** | **9.0248** | **0.3144** |
| | STL+SARIMA | 3.3000 | 2.3445 | 11.1895 | 0.1063 |
| | STL+XGBoost | 2.8006 | 2.1280 | 9.6849 | 0.2415 |
| | STL+LSTM | 3.1118 | 2.2856 | 10.5996 | 0.1572 |

The Skill Score metric, defined as:
$SS = 1 - \mathrm{RMSE}_{\mathrm{model}} / \mathrm{RMSE}_{\mathrm{naive}}$

where $\mathrm{RMSE}_{\mathrm{naive}}$ represents persistence forecasting ($Y_{t+h} = Y_t$), deserves specific attention. Negative Skill Score values, observed for SARIMA at $t + 1$ ($SS = -0.072$), STL+XGBoost at $t+1$ ($SS = -0.745$), and STL+LSTM at $t+1$ ($SS = -0.775$), indicate that these configurations fail to outperform the trivial naive forecast at the immediate horizon. This is a critical finding: the three STL hybrid models not only underperform their end-to-end counterparts, but actively regress below the forecasting baseline at a short horizon. The persistence of negative Skill Scores exclusively within Group B at $t+1$ provides further empirical evidence that the STL preprocessing step injects net noise into the prediction pipeline under the structural conditions of this dataset.

The end-to-end LSTM model consistently demonstrates superior performance across all evaluated forecasting horizons. Crucially, the empirical data yields a uniform negative result regarding the deployment of STL preprocessing: every hybrid model variation in Group B exhibits strict performance degradation compared to its Group A equivalent. At horizon $t + 1$, extracting the STL components causes the LSTM error profile to jump from 1.5450°C to 2.8482°C.

The predictive accuracy drops severely when the data undergoes explicit decomposition. To verify that this degradation is not an artifact of random sample variation, a Diebold-Mariano predictive accuracy test [5] was conducted. The complete DM test constitutes a definitive statistical result: all nine hypothesis tests reject $H_0$ at the 5% significance level, and six of nine achieve $p < 0.01$. The degradation effect is therefore neither horizon-specific nor model-family-specific — it is a systematic consequence of the structural mismatch between the STL pipeline assumptions (strong trend presence) and the empirical properties of this dataset ($FT = 0.014$). Notably, the strongest rejection occurs for the LSTM pair at $t + 1$ ($DM = -5.44$, $p < 0.001$), consistent with LSTM's superior capacity for end-to-end seasonal learning that is disrupted most severely by decomposition.

## V. Discussion and Theoretical Breakdown

The consistent failure of the STL hybrid models across all configurations contradicts the common assumption that decomposition simplifies the forecasting problem. This systematic drop in performance can be explained by three distinct structural mechanisms:

1. *The trend-induced noise injection mechanism:* Because the empirical strength of trend is nearly zero ($F_T = 0.0140$), the local Loess tracker captures random multi-year fluctuations rather than a persistent underlying signal. It is notable that even SARIMA, the model best equipped to handle seasonally integrated series, yields a marginally negative Skill Score at $t + 1$ ($SS = -0.072$). This reflects the well-documented limitation of SARIMA at immediate horizons: the persistence naive forecast ($Y_{t+1} = Y_t$) often outperforms autoregressive models over one-step intervals in low-noise conditions, since differencing introduces a moving-average component that slightly inflates one-step-ahead error variance. When this

unstable trend is extrapolated forward, it acts as additive noise, compounding errors during the final profile reconstruction.

2. *Context stripping and residual overfitting:* Machine learning models like XGBoost and deep neural networks like LSTM excel at learning complex seasonal shapes and temporal interactions directly from unadjusted data sequences. Isolating the seasonal pattern removes this essential context. Forcing these architectures to optimize exclusively on highly volatile remainder data ($R_t$) strips away the smooth seasonal boundaries, causing the models to overfit to short-term noise.

3. *Multi-step error propagation:* In the end-to-end setups, errors across extended horizons ($t+3 \rightarrow t+7$) increase gracefully. In contrast, the hybrid configurations suffer from compounding errors across three independent forecasting channels ($\hat{T}$, $S$, $\hat{R}$). This multi-component compounding degrades accuracy faster than direct modeling, as evidenced by the high MAE metrics.

4. *The comparative value of explicit seasonal modeling:* Beyond the primary STL analysis, the results surface a secondary finding of independent relevance: the statistical SARIMA model achieves superior accuracy over XGBoost at the extended t + 7 horizon (RMSE = 2.7136 vs. 3.7266, a 37.2% reduction), and remains competitive with LSTM at t+1 (RMSE = 1.7199 vs. 1.5450; DM p = 0.089, not significant at 5%). This pattern reflects a structural advantage of explicit seasonal parameterization. SARIMA's seasonal differencing operator (D = 1, s = 365) encodes the annual periodicity as an architectural constraint, the model cannot violate this structure regardless of the training window. XGBoost, by contrast, relies on lag-365 features to implicitly approximate the same periodicity; as the forecast horizon extends beyond the training distribution, this implicit representation degrades more rapidly than the constrained ARMA structure. The finding reinforces the canonical time series principle that domain knowledge encoded as structural model constraints, rather than learned from data alone, provides robustness benefits at longer horizons. For practitioners, this implies that SARIMA remains a competitive and interpretable baseline for multi-day temperature forecasting, particularly when computational resources for deep learning are limited.

Based on these findings, we establish a definitive architectural guardrail: our empirical findings suggest that STL preprocessing may become unreliable under extremely weak trend regimes (e.g., FT $\approx 0.014$ in this study). When $F_T \rightarrow 0$, end-to-end recursive structures should be preferred.

## VI. Conclusion

This study provides a rigorous empirical evaluation of the structural limits of decomposition-based forecasting. Using an extensive meteorological dataset from Hanoi, we demonstrated that applying STL decomposition to a time series with a weak trend ($F_T = 0.0140$) systematically degrades predictive performance across statistical, machine learning, and deep learning models. The end-to-end LSTM architecture achieved the highest accuracy across all horizons, outperforming the preprocessed alternative by avoiding the noise injection and overfitting inherent in the hybrid pipeline. Future research will focus on dynamically activating decomposition layers based on real-time variance calculations over rolling windows.

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